

O-RAN Open Fronthaul S-Plane — Master Test Case Catalogue

Validation Record and Coverage Statement · Revised 2026-09-17 · **validation figures superseded 2026-09-29 — see notice below**

SUPERSEDED IN PART — revised 2026-09-29. The 2026-09-17 **132-run** figures below are **withdrawn**. A cross-check audit on 2026-09-20 found four defects in the campaign apparatus, two of which fabricated data — including that B3_pdv_congestion never injected anything (this kernel has no sch_netem and the failure was swallowed), which **invalidates B3 in the 132-run campaign as well**. All scenarios were re-executed as a **168-run** campaign (12 replicates × 14 scenarios), recounted from the raw per-run files on 2026-09-29.

Current authoritative figures (168 live runs, per-scenario macro averages): pre-frozen base rule — attack sensitivity **0.625**, specificity **0.783**, attribution **0.552**; post-observation decision_rule_v3 (primary) — attack sensitivity **0.990**, specificity **0.783**, attribution **0.875**; abstention correctness **1.000**. Specificity is 0.783 because B_bc_replacement scores 0/12 (12 false ATTACK calls) and B3 abstains on 1 of 12; excluding B_bc_replacement it is **0.979**, not 1.000. C1_removal is 11/12: the interception detector fires below a 0.500 delivered/declared rate ratio, caught 65 % blackhole (ratio 0.458–0.462) and missed 60 % (ratio 0.505).

Authority for these numbers: splane_campaign_CORRECTED_2026-09-20.tgz (sha256 6149b4fb...), EVALUATION_V4.json, and ORAN_SPlane_AUDIT_AND_CORRECTIONS (2026-09-29 revision). The configuration, topology, standards traceability, fault taxonomy and test-case definitions in this document are **unchanged and remain valid**; only the validation figures are superseded.

STATUS AS ISSUED 2026-09-17 — NOW WITHDRAWN. Randomised 132-run campaign (12 reps, 11 scenarios): attack sensitivity **1.000 [0.940, 1.000]** (n=60, 60/60, zero false negatives); decidable-benign specificity **1.000** (48/48); honest abstention **12/12**; attribution 0.817. Frozen rule (18 hashed artefacts) applied to **real TIMESAFE hardware** attack captures **detects 5/6 by standards/protocol rules alone**. One documented limitation (B_bc_replacement, single-BC schema) reported not patched. Figures in the body below have been updated to this campaign; the prior 81-run revision is retained under **_pre_2026-09-17_doc_update/**.

O-RAN Open Fronthaul S-Plane
Master Test Case Catalogue, Validation Record and Coverage Statement

How to read this document

Status	Meaning
RUN	Implemented on our testbed and executed. Results reported in §7.
SPEC	Specified here and buildable in software, but not yet executed. Each says what it would catch.
IMPOSSIBLE	Cannot be produced or measured on a software testbed. Physical reason given.

On the question of exhaustiveness. Test coverage of an adversarial space cannot honestly be given as a percentage – there is no fixed denominator, because an attacker can always invent a new variation. Any figure such as "99.9% covered" would be an invented number. This document instead reports coverage over denominators that are real and countable: how many catalogued cases exist, how many are executed, how many are specified-but-unrun, and how many are physically impossible here. Those figures are checkable. §8 states explicitly what is not covered.

Primary source					
Attack cases	60	5	unrun)	(hardware)	
Research synthesis: TIMESAFE, PTPsec,			31	24	
Springer survey, O-RAN WG11, Concordia,					TUM,
MDPI, CVE records					
Conformance / negative	50+	6 (as legality	38	6	UNH-
IOL IEEE 1588 test plans; ITU-T		checks)			
G.8275.1 Amd.3; G.8273.2; G.8260					
Benign fault scenarios	58	6	39	15	ITU-T
G.827x; vendor app notes (Calnex,					
Microchip, Meinberg, Cisco, Nokia);					
operator incident reports					
TOTAL	168+	15	108	45	

Honest reading of these numbers. 19 of 168 catalogued cases are executed – roughly 11% (2026-09-17: +2 benign gap scenarios, +2 live conformance-legality checks); plus cross-validation on 6 real TIMESAFE hardware captures. That sounds low, and it is the truthful figure. But the 15 that run are not a random sample: they are the cases that (a) map to official O-RAN WG11 S-plane test IDs, and (b) are the ones the project's research claim actually rests on. Of the faults that are physically possible on a software testbed, 9 of 9 fault families are exercised. The large SPEC column is the honest measure of how much room remains.

2 · Standardised constants — replacing invented thresholds

The most valuable outcome of the literature survey. Each row replaced, or corroborated, a constant that had previously been chosen by us.			
Constant	Standardised value	Source	Effect on our
rule			
Foreign-master	FOREIGN_MASTER_THRESHOLD = IEEE 1588 §9.3.2.4; linuxptp		REPLACES our
invented "1% transient filter". Now			
qualification	2 Announce within a 4	foreign.h; behaviourally proven by	a real
protocol rule.			
	announce-interval window (= 0.5 s at UNH PWR.c.2.3		
	8/s)		
Message-interval	±30% of the stated mean, at 90%	IEEE 1588-2019 cl.7.7.2.1 (via Calnex	REPLACES an
ad-hoc rate band.			
tolerance	confidence	PFV)	
Successive-interval cap	Successive Sync/Announce intervals	ITU-T G.8275.1 Amd.3	CORROBORATES
our 2× cadence tolerance – it			
	must not exceed 2× the mean		was the right
factor, now with a citation.			
stepsRemoved discard	Announce with stepsRemoved ≥ 255	UNH PWR.c.2.4; G.8275.1	NEW check,
not previously implemented.			
	must be discarded	maxStepsRemoved {1-255}	
announceReceiptTimeou	3 – 3 × 1/8 s = 375 ms to declare	ITU-T G.8275.1 Amd.3; Cisco portDS	Confirms our
sub-2 s detection budget arithmetic.			
t	Announce loss		
Legal clockClass set	{6, 7, 13, 14, 135, 140, 150, 160, 165,	ITU-T G.8275.1 Amd.3; Calnex Annex	STRENGTHENS our
"clockClass ≥ 6" rule into an			
	248, 255} – anything else is a	A	explicit allow-
set.			
	conformance failure		
clockAccuracy	0x21 PRTC-locked; 0xFE non-PRTC	Calnex G.8275.1 Annex A	NEW legality
check available.			
offsetScaledLogVariance	0x4E5D PRTC; 0xFFFF other	Calnex G.8275.1 Annex A	NEW legality
check available.			
Transport restriction	Ethernet multicast only; VLAN tags	ITU-T G.8275.1; Calnex Annex A	NEW – a VLAN-
tagged PTP frame is itself			
	not allowed; unicast prohibited		non-
conformant.			
Constant	Standardised value	Source	Effect on our rule
PTSF states	PTSF-lossSync / lossAnnounce /	G.8275.1/.2 §6.7.11; linuxptp	NEW state-based evidence channel.
	lossDelayResp; port ignores		
	Announce from that source while		
	set		

3 · ATTACK test cases

60 catalogued. Families A-K. O-RAN column gives the official WG11 Security Test Specification ID where one exists (ETSI TS 104 105).

3.1 Announce / BMCA / rogue master

ID	Attack	Mechanism / field	Detector signature
O-RAN ID	Status		
A1	Rogue grandmaster	Announce with superior priority1/clockClass	Unknown GM identity wins BMCA;
11.1.5.2.1	RUN	/accuracy/variance/priority2	illegal clockClass; drift >40,000
ns			
A2	GM impersonation	Clone legitimate GM identity + MAC	Same identity from two
ports/MACs	11.1.5.2.1 SPEC		
A3	clockIdentity forcing	Crafted identity (e.g. embedded ffff) to win	Anomalous identity byte
pattern	11.1.5.2.2 SPEC		
		tiebreak	
A4	stepsRemoved	Advertise stepsRemoved=0 to appear	stepsRemoved inconsistent with
–	SPEC		
	manipulation	closer to PRTC	physical hop count
A5	BMCA oscillation /	Two near-equal crafted masters alternate	Rapid repeated GM/port-state
–	SPEC		
	flapping	wins	transitions; servo never
settles			
A6	Foreign-master table	Flood many spoofed clockIdentities to	Explosion of tracked foreign
masters;	– SPEC		
	exhaustion	overflow foreignMasterDS (min capacity 5)	election instability
A7	Announce-timeout	Suppress legit Announce until timeout, then	announceReceiptTimeout expires
then	11.1.5.3.1 SPEC		
	starvation	inject	rogue takeover
A8	alternateMasterFlag	Abuse optional alternate-master to inject	Sync accepted from a port that
is not	– SPEC		
	abuse	Sync from non-elected port	the elected GM

3.2 Timestamp forgery / correctionField / replay

ID	Attack	Mechanism / field	Detector signature
O-RAN ID	Status		
B1	preciseOriginTimestamp	Alter T1 in Follow_Up	Offset step with no matching
path-delay	– RUN (as		
	forgery		change
A2_sync			
_spoof)			
B2	originTimestamp forgery	Alter T1 in one-step Sync	Offset jump without delay
anomaly	– RUN		
B3	Coherent T1+T4 attack	Increment Follow_Up T1 and Delay_Resp	No delay anomaly – documented as
–	SPEC –	T4 together so delay math stays	"devious, hard to detect"
high		consistent	
value			
B5	twoStepFlag confusion	Flip twoStepFlag / send Follow_Up for a	Follow_Up presence
inconsistent with	24.2.1.2 SPEC		
		one-step Sync	flag
C1	correctionField tamper	Modify C2 in Follow_Up	Desync 100 µs–98 ms; negative
path	– SPEC		
	(fwd)		delays
C3	TC residence-time	Compromised transparent clock writes	correctionField inconsistent
with	– SPEC –		
	falsification	false residence	measured residence
needs a			
TC in			
topology			
C4	Boundary-clock	Compromised T-BC re-originates	Whole downstream branch
desyncs	11.1.5.2.2 RUN (as		
	compromise	manipulated timing downstream	coherently
A8_rogu			
e_bc)			
D1	Full replay	Retransmit captured Sync/Follow_Up	sequenceId regression; delay
spikes	– RUN (as		
			>20 M ns
A3_repla			
y)			
D2	Selective / partial replay	Replay only chosen messages	Intermittent stale sequenceId
–	SPEC		
D3	Delay_Req/Resp replay	Replay reverse-path exchange	Asymmetric delay estimate
–	SPEC		

3.3 Delay attacks — the hardest family

Status	ID	Attack	Mechanism	Signature
RTT asymmetry	E1	Static asymmetric delay IMPOSS	MitM adds constant delay one direction only	Persistent offset; $\alpha \neq 0$
IBLE to measure				
caught only by	E2	Incremental slow-ramp IMPOSS	Gradually growing one-way delay (e.g. 285 ns/s) to evade thresholds	Slow offset drift; cyclic asymmetry
IBLE to measure				
indicator	E3	Symmetric large-step delay IMPOSS	Equal both-direction delay; servo overshoots (~84 ms after 200 s)	Delay-variance
IBLE to measure				
oscillation	E4	Cyclic / periodic delay IMPOSS	Periodic modulation inside servo bandwidth, mimics diurnal jitter	Periodic offset
IBLE to measure				

Status	ID	Attack	Mechanism	Signature
SPEC (injectable)	E6	Flooding-induced queueing delay	Congest queues to add asymmetric delay (~18 μ s @ 83 Mbps)	Piecewise-linear delay; HW timestamping bypasses it
SPEC	E7	Link-speed asymmetry	Force asymmetric speed negotiation (100–10 Mbps), ~2 μ s fixed delay	Abnormal speed negotiation
SPEC	E9	Slave spoofing (early Delay_Req)	Inject Delay_Req with expected seq before the real slave	Duplicate/early Delay_Req

Why the E-family is marked impossible. sch_netem is unavailable in this kernel (tc ... netem → "Specified qdisc kind is unknown"), so these asymmetries cannot be injected here at all. What cannot be done is measuring the resulting time error: a passive capture yields only t1 and t4; the client's t2 and t3 never appear on the wire. Detection requires live per-client pmc telemetry (now logged but not yet evaluated) or a redundant second path. This is a measurement gap, not an injection gap.

3.4 DoS / resource exhaustion

Status	ID	Attack	Mechanism	O-RAN ID
11.1.5.1.1 / (A5)	F1	PTP message flooding RUN	High-rate PTP frames overwhelm nodes	24.2.1.1
11.1.5.3.1	F2	Complete packet removal SPEC —	Drop all PTP at an intermediate node → slave free-runs	
O-RAN mandate				
d				
11.1.5.3.1	F3	Selective Delay_Req removal SPEC	Drop only Delay_Req; delay estimate freezes	
SPEC	F4	ARP/MAC spoof blocking	Bind attacker MAC to O-RU IP so PTP never arrives	—
IMPOSS	F5	O-RU Crypto-engine exhaustion	Bogus MACsec frames exhaust receiver crypto CPU — crypto worsens DoS	—
IBLE (no MACsec)				
)				
IMPOSS	F6	Unicast-negotiation exhaustion	Flood REQUEST_UNICAST_TRANSMISSION; no	—

			requestor auth	
IBLE (G.				
8275.2				
only)				
	F7	L1 physical flooding	Physical fibre access, flood L1	—
IMPOSS				
IBLE (no				
PHY)				

3.5 Protocol-field abuse — the least-defended surface

These are trivially craftable, produce whole-second errors, and are almost absent from the attack literature. Our extractor already collects every field involved; none is yet exercised.

	ID	Attack	Field	Effect
Status				
	H1	versionPTP downgrade	versionPTP	Messages dropped; slave free-runs
SPEC				
	H2	Domain crossing	domainNumber	Cross-domain injection or forced
discard	SPEC			
	H3	currentUtcOffset manipulation	currentUtcOffset	Whole-second UTC jump, no servo
SPEC —				anomaly
novel				
	H4	leap61 / leap59 injection	leap flags	Spurious ±1 s step
SPEC —				
novel				
	H5	Traceability stripping	timeTraceable/frequenc	Tricks QL/holdover logic
SPEC —			yTraceable	
novel				
	H6	ptpTimescale confusion	ptpTimescale	Epoch misinterpretation (~37 s
TAI-UTC)	SPEC			
	H8	PATH_TRACE TLV abuse	PATH_TRACE	Hide timing loops / fake topology
SPEC				
	H10	AUTHENTICATION TLV	Annex P TLV	Force fallback to unauthenticated
PTP	SPEC			
	H11	strip/downgrade Management SET abuse	\$15 management msgs	In-band dataset reconfiguration
SPEC				
	H12	Malformed / fuzzed frames	Header & TLV lengths	Parser crash; daemon fault
SPEC —				
O-RAN				
24.2.1.2				
	H13	CVE-2021-3570 (linuxptp)	Forwarded msg length check	Daemon crash / info leak on a BC
SPEC				

3.6 Below-the-wire and cross-layer

	ID	Attack	Why it matters
Status			
	G1-G4	GNSS jam / spoof / meaconing / PRTC	Upstream of PTP entirely; receiver reports
healthy	IMPOSS		
		firmware compromise	while time is wrong
IBLE (no			
GNSS;			
RF			
illegal			
outside			
a chamb			
er)			
	I1-I3	Kernel-level time interception — hook	Defeats every packet-based detector and all
SPEC —			

Extension to section 3 (ATTACK test cases, pp. 2-5): the four columns per attack family

Added 5 Oct 2026. This page extends the table on the preceding page with four columns requested after the PRISM review: **Source** (who or what originates the attack or fault), **Destination** (where it hits in the O-RAN architecture), **Attack devices** (devices that could originate it; for benign faults, the originating devices) and **Consequence**. All pages before and after this one are unchanged from the 29 Sep 2026 revision. The catalogue holds 60 attack cases in families A-K; the columns are given per family because members of a family share source, destination, devices and consequence.

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
3.1 A1-A8	Announce / BMCA / rogue master	Clock or host sending manipulated Announce / BMCA attributes to win or destabilise master election [R7384 s3.2.4; TR106 T-SPLANE-02/03].	BMCA and foreign-master data set of every O-RU / clock on the segment; the O-RU's parent selection [TS982].	Compromised O-DU, BC, switch or O-RU, or an unauthorised host on the fronthaul Ethernet; supply-chain backdoor [TSAFE s4; TR106 T-FRHAUL-02].	False time to all downstream O-RUs; election instability / flapping [R7384 Table 1]; O-RU crash ~2 s after spoofing in one published experiment (switch ports in dynamic role) [TSAFE].
3.2 B1-B5, C1-C4, D1-D3	Timestamp forgery / correctionField / replay	Attacker altering or re-sending timing messages (T1/T4, correctionField, residence time, captured Sync/Follow_Up) [R7384 s3.2.1 manipulation, s3.2.2 spoofing, s3.2.3 replay].	The O-RU servo's offset and delay computation (timestamps and correctionField it consumes) [1588].	In-path element (switch, transparent clock, compromised T-BC) or injector on the segment [R7384 s3.1 MITM / injector; TSAFE s4.1].	False time or accuracy degradation; a coherent T1+T4 change can leave no delay anomaly [R7384 Table 1; MTC].
3.3 E1-E9	Delay attacks	In-path attacker delaying PTP messages, symmetrically or asymmetrically, constant, ramped or periodic [R7384 s3.2.6; TS105 11.1.5.3.2].	O-RU offset computation, which assumes a symmetric path [1588].	Any in-path forwarding element: switch, BC/TC, router; physical link access for speed manipulation [R7384 s3.1].	Time error of half the asymmetry, invisible to field checks; delay attacks are the hardest family to detect [1588; R7384 Table 1; MTC].
3.4 F1-F7	DoS / resource exhaustion	Flooding, removal of PTP packets, crypto/negotiation exhaustion, L1 flooding [R7384 s3.2.5, s3.2.7, s3.2.8, s3.2.9; TR106 T-SPLANE-01/04].	PTP processing of the master and O-RU ports; delivery of PTP between BC and O-RU [TR106 T-SPLANE-01/04].	Host or compromised device on the segment; in-path switch/BC for removal; physical fibre access for L1 [TSAFE s4; MTC].	Clock service interrupted, slave free-runs or degrades [TR106 T-SPLANE-01/04]; measured in C1: 36/36 RU ports lost master on Announce timeout [MEAS].
3.5 H1-H13	Protocol-field abuse	Crafted values in version, domain, UTC offset, leap, traceability, timescale, TLV, management or length fields; malformed frames [R7384 s3.2.2, s3.2.11; TS105 24.2.1.2].	O-RU PTP parser and time-properties data set; management interface [TS982; MTC].	Any host able to transmit on the segment, including a compromised O-RU [TSAFE s4].	Whole-second false time (UTC/leap abuse), protocol confusion, or parser crash / daemon DoS depending on implementation [R7384 Table 1, s3.2.11; MTC].
3.6 G1-G4	GNSS jam / spoof / meaconing / PRTC firmware	RF interference or counterfeit GNSS signals, or compromised PRTC firmware [R7384 s3.2.10; BSI].	PRTC / T-GM GNSS receiver: the root reference (and the O-RU receiver in LLS-C4) [R7384 s3.2.10].	RF jammer, GNSS simulator / spoofer near the antenna; compromised GM firmware [BSI].	Whole-network false time or holdover while PTP stays legal; receiver may report healthy [R7384 s3.2.10; MTC].
3.6 I1-I5	Kernel-level time interception / PHC exploits	Malicious code on a clock's host intercepting the OS time interfaces or exploiting PHC / adjtimex flaws [MTC].	Operating system time path of the O-RU / BC host itself.	Compromised O-RU, O-DU or BC host (root-level malware) [MTC].	Servo reports locked while time is wrong; defeats packet-based detection [MTC].
3.6 J1-J6	SyncE ESMC / VLAN / MACsec layer	Spoofed ESMC quality levels, timing loops, VLAN/MAC spoofing, MACsec bypass [MTC].	Frequency-source selection (SyncE) and the S-plane VLAN / link security.	Device with access to the physical link or S-plane VLAN; compromised switch [MTC].	Hijacked frequency source selection; unprotected VLAN enables most other families [MTC].
3.6 K1-K4	Topology-specific composites (LLS-C1..C4)	Attacks combining the families above for one synchronisation configuration [MTC].	LLS-C1/C2: O-DU -> O-RU link; LLS-C3: GM/BC chain; LLS-C4: O-RU local GNSS [TS982; Armstrong ATIS 2023].	Compromised O-DU (C1/C2), BC chain element (C3), RF attacker (C4) [TSAFE s4; BSI].	As the component families; scope depends on how many O-RUs share the compromised element.

Key to bracketed tags: **R7384** IETF RFC 7384 (2014): s3.1 attacker classes, s3.2 threats, Table 1 impact (false time / accuracy degradation / DoS) · **TR106** ETSI TR 104 106 V3.0.0 (2025-06) O-RAN threat model: s7.2 threat agents; T-SPLANE-01..04; T-FRHAUL-01/02 · **TS105** ETSI TS 104 105 V7.0.0 (2025-06) O-RAN security test specification (test-case IDs) · **TS982** ETSI TS 103 982 V8.0.0 O-RAN architecture: O-RU hosts Low-PHY + RF and terminates Open Fronthaul (cl. 3.1, 6.3.6) · **TSAFE** Groen et al., TIMESAFE, ACM TOPS 28(5) 2025 / arXiv 2412.13049v3: s4 access paths and attacks; O-RU crash ~2 s after attack start with switch PTP ports in dynamic role (one observed experiment, not a standardised limit) · **G8275** ITU-T G.8275.1 telecom profile clockClass semantics via Calnex one-page summary: 6 T-GM locked, 7 T-GM holdover in spec, 135 T-BC holdover in spec, 140/150/160 T-GM out of holdover spec, 165 T-BC out of holdover spec, 248 no reference since start-up · **G8271** ITU-T G.8271.1 network time-error budget: 1.5 us at reference point E, 1100 ns network limit at point C (v8 deck slide 7; Calnex budget summary) · **G8261** ITU-T G.8261 packet delay variation (PDV) context · **3GPP** 3GPP TS 38.133 cell phase sync < 3 us (via Ruffini, ATIS 2018); TS 38.104 cl. 9.6.3.2 TAE 65 ns MIMO · **BSI** BSI 5G Risk Analysis: Radio Access Network, v1.0 (2025) - GNSS jamming/spoofing · **1588** IEEE 1588-2019 offset/delay computation (asymmetry error = half the path asymmetry) · **MEAS** Measured in this project (168-run campaign, RU1-RU3 ptp4l logs, 36 logs per scenario) · **B6M** Measured in this project: two-laptop crystal comparison, 2 Oct 2026, outputs/B6_two_machine_2026-10-02/run2/DRIFT_REPORT_run2.md (run 2 ESTABLISHED, +21.013 ppm [20.623, 21.403]; run 1 NOT ESTABLISHED, +20.161 ppm; the 0.852 ppm run-to-run difference exceeds either CI). Consumer laptop crystals against an NTP reference - not telecom-class oscillators · **MTC** this catalogue's own entry for the family.

is wrong. novel,		clock_gettime / ADJ_FREQUENCY	crypto. The servo reports "locked" while time
high			Sharpest challenge to this entire approach.
value			
	I4-I5	PHC ioctl exploit (CVE-2025-21814), adjtimex	Kernel fault / privilege bypass
SPEC			
		leak (CVE-2018-11508)	
	J1-J3	SyncE ESMC QL spoofing, timing-loop	Hijacks frequency-source selection; explicitly
noted as	IMPOSS		
		induction, QL-DNU injection	uncovered by O-RAN WG11
IBLE (no			
PHY/ES			
MC)			
ID	Attack	Why it matters	Status
J4-J6	VLAN/MAC spoofing, MACsec bypass/downgrade	S-plane VLAN often unprotected; enables most other families	SPEC (VLAN) / IMPOSS
			IBLE (M
			ACsec)
K1-K4	LLS-C1 direct-link injection; LLS-C2/C3 BC-chain poisoning; LLS-C4 local-GNSS spoof; cross-domain handover desync	Topology-specific composites	K2 RUN; others S
			PEC/IM
			POSSIB
			LE
4 · CONFORMANCE and NEGATIVE test cases			
From the UNH-IOL IEEE 1588 conformance test plans and ITU-T G.8275.1 Amd.3. These are protocol-legality tests – Class A in our threshold taxonomy, meaning the standard states the exact pass criterion and no calibration is required.			
	UNH ID	Test	Pass criterion
Our			
status			
	c.1.1 / c.1.2	logAnnounceInterval /	Interval within ±30% of mean at 90% CI;
logMessageInterval	RUN (as	logSyncInterval	field correct
cadence			
legality)			
	c.1.3	announceReceiptTimeout	Value 3; must not claim master before timeout, must by
N	SPEC		intervals
			Defaults in Announce; G.8275.1 fixes priority1=128
RUN	c.1.5	priority1 / priority2	
			Announce in a non-matching domain must be ignored
SPEC –	c.1.7	domainNumber isolation	
high			
value			
negative			
test			
	c.2.1	Disqualify by own clockIdentity	Announce whose sourcePortIdentity matches DUT's own
SPEC			identity must be rejected (loop protection)
			Only the latest Announce per source affects selection
	c.2.2	Disqualify by most recent	
SPEC			≥2 Announce within 4 intervals to qualify; <2 must be
ignored;	c.2.3	Foreign-master window	foreignMasterDS ≥5 records
ADOPT			
ED into			
rule			
	c.2.4	stepsRemoved limit	≥255 must be discarded; 254 accepted
SPEC			
	c.2.5	alternateMasterFlag	Announce with flag TRUE must be discarded
SPEC			
	c.2.6	Dataset comparison	Ordering: priority1 → clockClass → clockAccuracy →
RUN (im			offsetScaledLogVariance → priority2 →
grandmasterIdentity		plicitly)	
	c.2.8	State decision algorithm	Correct BMC_MASTER / BMC_PASSIVE / BMC_SLAVE
SPEC			
			outputs
	c.2.9	stepsRemoved ±1 rule	If received ≥ own+1 → yield; ≤ own-1 → continue
announcing SPEC			
	c.7.1	Mean path delay	Correct computation from Delay_Req/Delay_Resp

Partially

RUN				
SPEC	—	PTSF states	lossSync / lossAnnounce / lossDelayResp; port ignores	
			Announce from a failed source	
unicast	—	Transport legality	Ethernet multicast only; VLAN tags not allowed;	
(mcast) /	RUN		prohibited	
SPEC				
(VLAN)				

5 · BENIGN fault scenarios — the look-alikes

58 catalogued. These are the events that resemble attacks but are not, and they are where a detector actually fails in deployment. Our benign coverage is the weakest part of the work.

ID	Benign scenario	Observable change	Mistakable for
for master	Status B-01 Planned GM failover RUN	New grandmasterIdentity; brief	A1 rogue
DoS / critical	B-02 Unplanned GM failover SPEC —	UNCALIBRATED-SLAVE; no clockClass degradation Announce timeout → BMCA reselect; clockClass may pass 7-52	Announce suppression
gap	B-03 GM cold reboot injecting degraded SPEC	clockClass walks 248-52-7-6 monotonically over minutes	Spoofers
manipulation attack	B-05 GM priority reconfiguration SPEC	Attribute change, no clockClass change;	Announce Priority-
insertion	B-06 New GM added to domain SPEC	deterministic domain-wide reselect New clockIdentity; commissioning Announce burst	Rogue GM
Intermittent	B-08 Dual-GM BMCA flapping rogue-master SPEC —	Repeated GM flips between two known GMs; O-RAN calls this "S-1 Grandmaster flapping"	attack
high value	B-09 GM clockIdentity change after impersonation SPEC	"New" GM at same topological position, same attributes	GM
attack	B-10 GNSS loss → reselect to alternate IMPOSS GM	Phase step equal to known cTE difference between GM paths; reproducible	Time-shift
IBLE (no GNSS)	B-11 Commissioning Announce burst flood DoS SPEC	Rate spike correlated with site power restore, decays monotonically	Announce-

ID	Benign scenario	Observable change	Mistakable for	Status
—	PDV / congestion	meanPathDelay variance, load-correlated, no protocol anomaly	Delay attack	RUN
—	Topology / BC reconfiguration	stepsRemoved change consistent with new hop count	Rogue BC insertion	RUN
—	Legitimate BC replacement	New BC identity in path, provisioned	A8 rogue BC	SPEC — critical gap
—	Real leap-second event	Genuine leap61 flag and 1 s step	H4 leap injection	SPEC
—	SyncE / EEC degradation, oscillator drift, holdover	QL downgrade; drift within mask	GNSS jam	IMPOSS IBLE (no PHY / os cillator)

6 · What we actually built and ran

Reps	Scenario	O-RAN / source ID	How produced on the wire
9	baseline	—	All nodes nominal, GM-A → BC → 3 O-RU clients
	A1_rogue_master list, self-announcing	11.1.5.2.1 9	ptp4l instance, priority2=1, identity off the allow-

Extension to section 5 (BENIGN fault scenarios, p. 7): the four columns per scenario

Added 5 Oct 2026. This page extends the table on the preceding page with four columns requested after the PRISM review: **Source** (who or what originates the attack or fault), **Destination** (where it hits in the O-RAN architecture), **Attack devices** (devices that could originate it; for benign faults, the originating devices) and **Consequence**. All pages before and after this one are unchanged from the 29 Sep 2026 revision. For benign scenarios there is no attacker; 'Attack devices' lists the originating devices.

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
B-01	Planned GM failover	Operator maintenance; primary GM withdrawn.	BMCA of every clock; O-RU parent.	GM-A / GM-B (provisioned).	Brief re-parenting; time stays traceable [MEAS: 12/12 BENIGN as B2].
B-02	Unplanned GM failover	GM failure, no maintenance window.	BMCA of every clock; O-RU parent.	Failed GM; provisioned backup GM.	Re-parenting; intent not knowable -> escalate [MEAS: 12/12 UNKNOWN].
B-03	GM cold reboot	GM restart (power or software).	Grandmaster data set seen by all clocks.	GM and its GNSS receiver.	clockClass walks back to 6 over minutes; transient degradation [G8275; MTC].
B-05	GM priority reconfiguration	Operator changes priority attributes.	BMCA domain-wide.	GM (management action).	Deterministic reselection; no time error if new GM is valid [MTC].
B-06	New GM added to domain	Commissioning of a new GM.	BMCA; foreign-master tables.	New provisioned GM.	New identity appears; benign only if provisioned [MTC].
B-08	Dual-GM BMCA flapping	Two known GMs alternating (marginal attributes).	BMCA and O-RU parent.	Both provisioned GMs.	Repeated GM flips between two known GMs; repeated re-parenting [MTC].
B-09	GM identity change after card swap	Hardware replacement in the GM.	Grandmaster identity seen by all clocks.	Replaced GM card.	'New' GM at the same position; benign only if provisioned [MTC].
B-10	GNSS loss -> reselect to alternate GM	GNSS outage at one GM.	BMCA; O-RU parent.	GM GNSS receiver.	Phase step equal to the cTE difference between GM paths [MTC].
B-11	Commissioning Announce burst	Site power restore.	O-RU PTP receive path.	GMs/BCs restarting.	Short rate spike that decays; not a flood [MTC].
-	PDV / congestion	Competing data traffic in the fronthaul [G8261].	BC -> O-RU PTP delivery.	Congested switch queue / link.	PDV 0.07 ms -> 5.6-857 ms measured; 12/36 RU logs Announce timeout [MEAS].
-	Topology / BC reconfiguration	Planned link re-homing.	Affected O-RU port state and path.	Reconfigured link / switch port.	Re-parenting with stepsRemoved consistent with the new hop count [MEAS: 12/12 BENIGN as B7].
-	Legitimate BC replacement	Planned BC swap, replacement provisioned.	O-RU parent selection.	Old BC, replacement BC.	Brief loss; mistaken for A8 by the base rule [MEAS: 0/12, 36/36 Announce timeout].
-	Real leap-second event	Scheduled UTC leap second.	Time-properties data set of all clocks.	GM (via GNSS / operator).	Genuine 1 s step in UTC; must not be flagged as H4 [MTC].
-	SyncE / EEC degradation, oscillator drift, holdover	Frequency reference or local oscillator degradation.	Node equipment clocks / local oscillators.	PHY/EEC, crystal/OCXO, GM GNSS receiver.	Drift within the declared mask [MTC; G8275]. Separately measured: two consumer laptop crystals differ by +21.013 ppm (not telecom-class) [B6M].

Key to bracketed tags: **R7384** IETF RFC 7384 (2014): s3.1 attacker classes, s3.2 threats, Table 1 impact (false time / accuracy degradation / DoS) · **TR106** ETSI TR 104 106 V3.0.0 (2025-06) O-RAN threat model: s7.2 threat agents; T-SPLANE-01..04; T-FRHAUL-01/02 · **TS105** ETSI TS 104 105 V7.0.0 (2025-06) O-RAN security test specification (test-case IDs) · **TS982** ETSI TS 103 982 V8.0.0 O-RAN architecture: O-RU hosts Low-PHY + RF and terminates Open Fronthaul (cl. 3.1, 6.3.6) · **TSAFE** Groen et al., TIMESAFE, ACM TOPS 28(5) 2025 / arXiv 2412.13049v3: s4 access paths and attacks; O-RU crash ~2 s after attack start with switch PTP ports in dynamic role (one observed experiment, not a standardised limit) · **G8275** ITU-T G.8275.1 telecom profile clockClass semantics via Calnex one-page summary: 6 T-GM locked, 7 T-GM holdover in spec, 135 T-BC holdover in spec, 140/150/160 T-GM out of holdover spec, 165 T-BC out of holdover spec, 248 no reference since start-up · **G8271** ITU-T G.8271.1 network time-error budget: 1.5 us at reference point E, 1100 ns network limit at point C (v8 deck slide 7; Calnex budget summary) · **G8261** ITU-T G.8261 packet delay variation (PDV) context · **3GPP** 3GPP TS 38.133 cell phase sync < 3 us (via Ruffini, ATIS 2018); TS 38.104 cl. 9.6.3.2 TAE 65 ns MIMO · **BSI** BSI 5G Risk Analysis: Radio Access Network, v1.0 (2025) - GNSS jamming/spoofing · **1588** IEEE 1588-2019 offset/delay computation (asymmetry error = half the path asymmetry) · **MEAS** Measured in this project (168-run campaign, RU1-RU3 ptp4 logs, 36 logs per scenario) · **B6M** Measured in this project: two-laptop crystal comparison, 2 Oct 2026, outputs/B6_two_machine_2026-10-02/run2/DRIFT_REPORT_run2.md (run 2 ESTABLISHED, +21.013 ppm [20.623, 21.403]; run 1 NOT ESTABLISHED, +20.161 ppm; the 0.852 ppm run-to-run difference exceeds either CI). Consumer laptop crystals against an NTP reference - not telecom-class oscillators · **MTC** this catalogue's own entry for the family.

A2_sync_spoof never sends	9	B1/B2 family	scapy-forged Sync with implausible originTimestamp;
A3_replay		D1	Announce Live slice captured then retransmitted with tcpreplay
A5_dos_flood		11.1.5.1.1 / 24.2.1.1	scapy high-rate PTP flood
A8_rogue_bc relaying downstream with	9	11.1.5.2.2	Two-port ptp4l (priority2=115) slaving upstream,
B2_gm_failover list,	9	B-01	stepsRemoved incremented GM-A withdrawn; BMCA elects GM-B, which IS on the allow-
B3_pdv_congestion		—	maintenance window open tbtf bottleneck BC→RU + competing non-PTP traffic
B7_topology_change window	9	—	RU3 link bounced and re-homed during an open maintenance

7 · Validation results — SUPERSEDED (2026-09-17 132-run campaign; withdrawn 2026-09-20, see notice on page 1)

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.940, 1.000]	60
Specificity (decidable-benign: baseline,B2,B3,B7)	1.000	[0.926, 1.000]	48
Abstention correctness (ambiguous benign→UNKNOWN)	1.000	[0.757, 1.000]	12
Fault-attribution accuracy	0.817	[0.701, 0.894]	60
Confusion – attack: TP 60, FN 0, UNKNOWN 0. Binary attack detection 60/60. Per-fault attribution: A2 12/12 · A3 12/12 · A5 12/12 · A1 8/12 · A8 5/12.			
Method: rule frozen (18 SHA-256 hashed artefacts) before any admissible run; all defect-exposing reps excluded and deleted; replicates 30-41 produced after the final freeze with the rule untouched. Every replicate uses per-rep randomised attacker identities/priorities/timings/bursts (run/randparams.py, seeded by rep), so the 132 runs are genuinely independent and the CIs are honest, not optimistic. Zero infrastructure failures across the campaign.			

All 12 specificity false positives are the single B_bc_replacement scenario: the frozen context schema holds one expected_bc_identity, so a PROVISIONED BC swap cannot be expressed and is flagged as a rogue BC (A8). This is reported as a limitation, NOT patched into the frozen result. Attribution confusion A1→A3 (x4) and A8→A3 (x7) is the documented re-parenting effect: inserting a rogue master/BC causes real sequenceId restarts that the A3 clause catches first; the binary ATTACK verdict is unaffected (60/60). The two new benign scenarios: B_unplanned_failover → UNKNOWN every time (correct honest abstention: a GM crash with no maintenance window is packet-indistinguishable from a suppression attack), B_bc_replacement as above.

Cross-domain validation (NEW): the frozen rule, applied to the REAL TIMESAFE hardware attack captures it was never built on, detects 5/6 by standards/protocol rules ALONE (no allow-list, nothing fitted), each citing its exact IEEE 1588 / G.8275.1 clause; the 1/6 miss is a healthy-looking spoof (conformant fields, no sequence regression) that requires provisioned context – the theoretically-predicted hard case.

8 · What these numbers do NOT establish

Limitation	Detail
Runs are now independent (RESOLVED) CI [0.566,1.000]). FIXED 2026-09-17:	Prior revision had zero randomisation (5 designs x 9 identical reps, effective n=5, per-rep randomised attacker params (run/randparams.py). 132 genuinely independent
runs, real n=60 per attack class;	sensitivity CI lower bound is now 0.940, not an optimistic artefact.
Some benign runs are easy (partly crash, no maintenance → the rule addressed 2026-09-17) exposes the single-BC schema	Two hard benign gap scenarios were added 2026-09-17: B_unplanned_failover (GM correctly abstains, UNKNOWN 12/12) and B_bc_replacement (a provisioned BC swap →
(baseline/PDV/topology) still inflate	limitation). The A1-vs-B2 discrimination remains the core claim; the easy benigns
separately from the abstention axis (n=12).	raw benign counts, so specificity is reported over decidable-benign (n=48)
Self-designed attacks easier test than an independent	The detector was tested only against attacks written by the same author. That is an adversary.
Provisioned context is handed to configuration, but it narrows the claim to the rule	The allow-list tells the rule which clocks are legitimate. This is genuine operator "detect unprovisioned clocks" rather than "detect attacks from traffic alone".

9% catalogue coverage remaining work.	19 of 168 catalogued cases executed. The 108 SPEC cases are the honest measure of
No hardware-dependent fault measurement remain untested for tested	GNSS spoof/jam, GNSS holdover, SyncE/EEC, oscillator drift, and all delay-attack physical reasons stated in §3.3 and §3.6.

9 • Highest-value next tests

Priority	Test	Why
1	Randomise attack parameters per replicate identity-based checks (A1/A2/A8); the	Makes the CIs mean what they state. Expected to hold for
2	Unplanned GM failover (B-02) + legitimate BC Current specificity is untested against replacement	rate-based A5 check is where it may break. The two benign cases that genuinely look like attacks. them.
3	Packet removal / selective interception (F2/F3) all.	O-RAN mandated test 11.1.5.3.1 – currently not covered at
4	Malformed / fuzzed input (H12) itself.	O-RAN mandated test 24.2.1.2 – robustness of the parser
5	Whole-second field abuse (H3/H4/H5) from the literature, and every field is already	Trivial to craft, produce 1 s errors, essentially absent collected by our extractor.
6	Conformance negatives (c.1.7, c.2.4, c.2.5) exact standardised pass criteria,	Wrong-domain, stepsRemoved ≥255, alternateMasterFlag – no calibration needed.
Priority	Test	Why
7	Coherent T1+T4 attack (B3) intact. Directly probes the known	Documented as hard to detect because it leaves delay math blind spot.
8	Kernel-level time interception (I1-I3) sharpest challenge to this entire	Defeats every packet-based detector and all cryptography. The approach.

10 • Sources

Source	Used for
O-RAN WG11 Security Test Specification – ETSI TS 11.1.5.3.1/.2, 24.2.1.1/.2 104 105 V7.0.0	Official S-plane security test IDs 11.1.5.1.1/.2, 11.1.5.2.1/.2,
O-RAN WG11 Threat Modeling – ETSI TR 104 106 V3.0.0	Threat identifiers T-SPLANE-01..04
UNH-IOL IEEE 1588 Default PTP Profiles window behavioural proof Conformance Test Plan v0.0.2 IEEE 1588-2019	Numbered conformance tests c.1.x, c.2.x, c.7.x; foreign-master
clockClass; cl.11.3 Delay_Resp	cl.7.7.2.1 ±30% interval tolerance; §9.3.2.4 foreign master; Table 5
ITU-T G.8275.1 + Amd.3 successive-interval cap	sequenceId; §15 management; §16 TLVs; Annex P
ITU-T G.8271.1 / G.8272 / G.8273.2 / G.8260 / G.8261	Profile values, alternate BMCA, clockClass ladder, maxStepsRemoved,
wander, SyncE / G.8262 / G.8264	
TIMESAFE – ACM ToPS 2025, doi:10.1145/3775060 and cell drop needing a manual reboot, switch PTP ports in dynamic role (preprint arXiv:2412.13049)	Fronthaul attack behaviour; real hardware captures; ~2 s O-RU crash
PTPsec – arXiv:2401.10664	Cyclic path-asymmetry analysis for delay-attack detection
Springer Cybersecurity s42400-021-00080-y	PTP attack taxonomy; why delay attacks defeat MACsec/Annex P
Concordia NCC – S-plane in O-RAN: Overview, Security, Research Directions	Threats T1-T8; LLS-C topology-specific attacks
TUM – Feasible Time Delay Attacks Against PTP MDPI Electronics 9/9/1398	Flooding-induced and link-speed asymmetry delay attacks
arXiv:2510.06421 – Breaking Precision Time	Coherent T1+T4 time-reference attack; correctionField tampering
CVE-2021-3570 / USN-6097-1	Kernel/OS-level time interception; CVE-2025-21814, CVE-2018-11508
linuxptp v4.0 source (foreign.h, ds.h, tlv.h)	linuxptp forwarding length-check vulnerability
	FOREIGN_MASTER_THRESHOLD=2; pmc datasets; PORT_STATS_NP / PORT_SERVICE_STATS_NP
Calnex / Microchip / Meinberg / Cisco / Nokia operator practice application notes	Profile restatements, clockClass semantics, benign failure modes,

Full catalogues (60 attack cases, 50+ conformance tests, 58 benign scenarios) with complete URLs are provided as accompanying markdown files. This document is the systematic summary.